

Statistical and Machine Learning

Neural Networks for Sequences

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Question 1: Architectures for Sequences

The lecture introduced three families of RNN architectures, distinguished by the mapping between input and output: **vec2seq**, **seq2vec**, and **seq2seq** (aligned and unaligned).

- (a) For each of the following tasks, identify which architecture is most appropriate, and briefly justify your answer:
 - i. Predicting the next word given the previous words (unconditional language generation).
 - ii. Classifying whether a movie review is positive or negative.
 - iii. Part-of-speech tagging (assigning a grammatical tag to every word in a sentence).
 - iv. Machine translation from English to French.
- (b) For the unaligned seq2seq (encoder-decoder) case, describe the role of the context vector $\mathbf{c}$, and explain why it can become a bottleneck for long input sequences.

Question 2: Training RNNs – Teacher Forcing and BPTT

- (a) Explain what is meant by **teacher forcing** when training a language model, and describe one problem it causes at test time. What is the standard fix, and how does it work?
- (b) Explain why computing the exact gradient via backpropagation through time (BPTT) requires $O(T^2)$ time for a sequence of length T , and how **truncated BPTT** addresses this.
- (c) Explain the vanishing/exploding gradient problem in RNNs. Why does it occur, and what are two ways of addressing it?

Question 3: Gating Mechanisms – GRU and LSTM

- (a) In a GRU, describe the roles of the reset gate $\mathbf{R}_t$ and the update gate $\mathbf{Z}_t$. What does it mean, in terms of model behaviour, if $\mathbf{R}_t \approx 0$? What does it mean if $\mathbf{Z}_t \approx 1$?
- (b) LSTMs augment the hidden state with a separate memory cell $\mathbf{C}_t$. Write down the update equation for $\mathbf{C}_t$, and explain how the forget and input gates allow the model to retain information over very long time spans.
- (c) Both GRUs and LSTMs rely on *additive* (rather than purely multiplicative) updates to their state. Explain why this design choice helps mitigate the vanishing gradient problem, drawing an analogy with another architectural idea seen earlier in the course.

Question 4: 1D CNNs for Sequences

- (a) Explain how a 1D convolution over a sequence produces a feature map, and how max-pooling over time is used to obtain a fixed-length representation for sequence classification.

- (b) 1D CNNs can also be used to *generate* sequences, but must first be converted into **causal** convolutions. Explain what makes a convolution “causal”, and why this is necessary for generation but not for classification.
- (c) Compare 1D CNNs to RNNs as an approach to sequence modelling: what is one practical advantage of CNNs over RNNs, and what kind of information can RNNs capture that a “vanilla” 1D CNN (with a fixed receptive field) cannot?

Question 5: The Attention Mechanism

- (a) Define what is meant by queries, keys, and values in the context of attention, and write down the general formula for computing an attention output $\mathbf{z}_j$ as a function of $\mathbf{q}_j$ and the keys/values $(\mathbf{k}_i, \mathbf{v}_i)$.
- (b) Scaled dot-product attention defines the score as $a(\mathbf{q}, \mathbf{k}) = \mathbf{q}^\top \mathbf{k} / \sqrt{d}$. Explain why we divide by $\sqrt{d}$, referring to the statistical properties of $\mathbf{q}$ and $\mathbf{k}$.
- (c) Explain the connection between attention and Nadaraya-Watson kernel regression. In particular, what plays the role of the “query”, the “keys”, and the “values” in kernel regression?

Question 6: Transformers and Language Models

- (a) Self-attention allows all outputs to be computed in parallel during training, unlike RNNs. Explain why this is possible, and why an RNN cannot be parallelised in the same way.
- (b) Since self-attention is permutation invariant, transformers use **positional encodings**. Explain what problem this solves, and describe one desirable property of the sinusoidal positional encoding scheme discussed in the lecture.
- (c) Compare BERT and GPT as approaches to language modelling:
 - i. Which model is causal (autoregressive), and which is non-causal?
 - ii. What pretraining objective does each use?